

Decision Trees

Non-parametric Methods

The core-idea is to split the data space into regions via binary questions on data dimensions.

suppose $D = \{(x_i, y_i)\}$ i.i.d P_{xy}
 $x \in \mathbb{R}^d$ $y \in \{1, 2, \dots, K\}$

At any given node m , which represents a Region $R_m \subset \mathbb{R}^d$, containing N_m samples, the tree-algorithm asks a binary question as follows:

Suppose x^j denote the j^{th} dimension of x .
 $\& t$ be a scalar threshold.

Split : if $x^j \leq t$, go left
 $x^j > t$, go right

The optimization objective for splitting
At every node, search over every data dim.
 $\&$ threshold (j, t) , that minimizes the node "impurity".

Let $I(R)$ be the impurity function.

$$L(j, t) = \frac{N_L}{N_m} I(R_L) + \frac{N_r}{N_m} I(R_r)$$

where N_L, N_r are number of samples that traverse the left & Right regions R_L & R_r , respectively.

Example Impurity functions.

i) Entropy or Information gain.

suppose p_k is the proportion of training samples from class - k in region R_m .

Entropy of a Node

$$H(R_m) = - \sum_{k=1}^K p_k \log_2(p_k)$$

$$\hat{I}(j, t) = H(R_{\text{parent}}) - \left[\frac{N_L}{N_m} H(R_L) + H(R_r) \frac{N_r}{N_m} \right]$$

choose a split that minimize $-\hat{I}(j, t)$.

b) GINI Impurity:

measures the probability of incorrectly classifying a randomly chosen element if it were randomly labelled.

prob. of choosing a datapoint from class k is p_k .

prob. of incorrectly labelling is $(1-p_k)$

$$\text{Gini}(R_m) = \sum_{k=1}^K p_k (1-p_k)$$

choose the split (j, t) that minimizes the Gini impurity.

Regression Trees.

If $y \in \mathbb{R}$ (continuous), discrete class prob. based metrics can't be used.

If a sample is falling in region R_m , define the prediction within R_m

as

$$\hat{y}_m = \frac{1}{N_m} \sum_{i \in R_m} y_i$$

Impurity metric for the Regression Case

$$I(R_m) = \frac{1}{N_m} \sum_{i \in R_m} (y_i - \hat{y}_m)^2$$

i) How to choose candidate thresholds for a split?

For a node m containing N_m samples, extract all the N_m values for the feature x^i ,

sort the values, remove duplicates,

rank-order them $u_1 < u_2 < u_3 \dots u_p$

$$p \leq N_m$$

Define t_i for $i = 1 \dots p-1$

$$t_i = \frac{u_i + u_{i+1}}{2}$$

ii) Stopping Criteria / pruning.

a) maximum Depth ,

Define $D_{\max}$ as the maximum possible depth (h_{param}).

if D is the depth of a given node,
algo stops if $D = D_{\max}$.

b) minimum samples for a split

let N_m be the ~~no~~ of samples in
the current node, the algo. will
split further only if

$$N_m \geq N_{\min}.$$

c) minimum Impurity Decrease

split only if $I(j, t) > \epsilon$.